

# Bharatiya Antariksh Hackathon 2026

ISRO × Hack2skill — 3rd Edition · National Level

**Master Brief & Strategy Pack** for Team **Syntax Error**. Every problem statement, the official idea-submission rules, a scored shortlist tailored to your team, and a phase-by-phase plan. Prepared 25 June 2026.

NOTE: This PDF is the text master. The companion file **ISRO\_BAH2026\_Master\_Brief.html** shows every official problem-statement image inline — open it and press Ctrl+P -> Save as PDF for an image-embedded PDF.

## 1 · Snapshot — the facts that matter

|                                                         |                                                 |
|---------------------------------------------------------|-------------------------------------------------|
| 3–4 members (you have 4)                                | Students only: UG/PG/PhD, India, Free           |
| 15 problem statements                                   | 30-hour on-site grand finale                    |
| Idea phase = 10-slide concept deck (mandatory template) | Judged on ideation & approach — NOT a prototype |

**Rewards:** ISRO recognition, potential ISRO internship, ISRO-scientist mentorship for shortlisted teams, II-AC travel reimbursed for finalists. In 2025 ~140 teams were shortlisted from the idea phase, then narrowed to 30 finalists.

## 2 · Timeline & runway (2026)

|                     |                                                           |
|---------------------|-----------------------------------------------------------|
| 10 Jun              | Registration & idea submission opens (you are registered) |
| 15–16 Jun           | Problem-statement explainer sessions 1 & 2                |
| 1 Jul, 11:59 PM IST | IDEA SUBMISSION CLOSES — our gate (6 days from now)       |
| 20 Jul              | Final shortlist announced                                 |
| 21 Jul              | Induction session for selected teams                      |
| 6–7 Aug             | Grand Finale (30-hr on-site build)                        |

Read: in 6 days you only need a brilliant deck. If shortlisted (20 Jul), the induction->finale window (21 Jul->6 Aug) is ~16 days to pre-build data pipelines, baseline models and a dashboard skeleton so the 30 finale hours go to the wow-factor. With your work ethic, an ambitious PS is the right risk — the only thing to manage is data readiness.

## 3 · Mandatory idea-submission template (10 slides)

| #  | Slide                   | What judges want                                            |
|----|-------------------------|-------------------------------------------------------------|
| 1  | Title                   | Team name, leader, chosen Problem Statement                 |
| 2  | Team Members            | Leader + 3 members, each Name + College                     |
| 3  | Opportunity             | How different from existing ideas? / How it solves it / USP |
| 4  | Features                | List of features + a visual                                 |
| 5  | Process flow / Use-case | A clean flow or use-case diagram                            |
| 6  | Wireframes / Mockups    | Optional — a dashboard mock is a strong edge                |
| 7  | Architecture            | End-to-end architecture diagram                             |
| 8  | Technologies            | Concrete stack (models, libraries, data, deploy)            |
| 9  | Estimated cost          | Optional — shows maturity                                   |
| 10 | Thank You               | Close                                                       |

Slides 3, 5, 6, 7 reward exactly what your team is good at — product framing, clean architecture, dashboard mock.  
We over-deliver there.

## 4 · Scored shortlist — all 15 ranked for us

Weighted for Team Syntax Error: Dataset readiness 30% (your make-or-break concern), Team fit 25% (full-stack + AI + product), USP & demo 20%, 30-hr feasibility 15%, ISRO/national impact 10%. Out of 5.

| #  | PS   | Title (short)                                | Data | Fit | USP | Feas | Imp | Score |
|----|------|----------------------------------------------|------|-----|-----|------|-----|-------|
| 1  | PS4  | Route Resilience: roads + graph criticality  | 4.5  | 5   | 5   | 5    | 4.5 | 4.80  |
| 2  | PS12 | Satellite frame interpolation (optical flow) | 5    | 4.5 | 4   | 4.5  | 4   | 4.50  |
| 3  | PS5  | AI Digital Twin of India climate             | 4    | 5   | 4.5 | 4    | 5   | 4.45  |
| 4  | PS3  | Surface AQI & HCHO hotspots                  | 4.5  | 4   | 3.5 | 4    | 5   | 4.15  |
| 5  | PS1  | Urban heat mitigation (AI/ML)                | 5    | 4   | 3   | 4    | 4   | 4.10  |
| 6  | PS13 | Air-gapped MPLS predictive copilot           | 4    | 3.5 | 5   | 3    | 4   | 3.93  |
| 6  | PS14 | Forecast GEO particle radiation              | 4.5  | 3.5 | 3.5 | 4    | 4   | 3.93  |
| 8  | PS7  | Exoplanet detection (light curves)           | 4.5  | 3.5 | 3.5 | 4    | 3.5 | 3.88  |
| 9  | PS10 | Infrared image colorization                  | 4.5  | 3.5 | 3.5 | 3.5  | 3.5 | 3.80  |
| 10 | PS2  | Generative-AI cloud removal (LISS-IV)        | 3.5  | 4   | 4   | 3.5  | 4   | 3.78  |
| 11 | PS6  | Crop type + moisture + irrigation            | 3.5  | 4   | 3.5 | 3.5  | 4.5 | 3.73  |
| 12 | PS15 | Solar flare nowcast (Aditya-L1)              | 3.5  | 3.5 | 4   | 3.5  | 4.5 | 3.70  |
| 13 | PS11 | Cross-modal satellite retrieval              | 3.5  | 3.5 | 3.5 | 3.5  | 3.5 | 3.50  |
| 14 | PS8  | Lunar subsurface ice (Chandrayaan-2)         | 2.5  | 3   | 4.5 | 3    | 5   | 3.35  |
| 15 | PS9  | Wavefront reconstruction (SH-WFS)            | 3.5  | 2.5 | 3.5 | 3    | 3.5 | 3.18  |

Why the bottom rows score low for us: PS8 (lunar) & PS15 (solar) are most prestigious but data is gated (Chandrayaan-2 only at finale; Aditya-L1 via niche ISSDC-PRADAN) — your data risk. PS9 needs C-level real-time AO optimization, far from product strengths.

## 5 · Top 3 — deep dive (pick one)

### [1st] GOLD PS4 — Route Resilience: Occlusion-Robust Road Extraction & Graph-Theoretic Criticality

Data: open & ready | Killer dashboard demo | Built for a 30-hr hackathon | Strong defensible USP

**Idea in one line:** "See-through-occlusions" road extraction -> heal into a routable graph -> find single points of failure -> planner clicks a junction to simulate a flood/closure and instantly sees rerouting & travel-time blow-up.

- Why it wins: the PS is engineered for a 30-hr build and hands you a novel USP (Resilience Index, Gatekeeper Nodes, node-ablation what-if).
- Why it fits you: CV model + graph logic + interactive Streamlit/Leaflet dashboard = your full-stack + AI + product wheelhouse.
- Data (high): SpaceNet/DeepGlobe/OpenSatMap pretraining, OSM auto-ground-truth, Sentinel-2 & Resourcesat LISS-IV open via Bhoonidhi; Cartosat-3 at finale.
- Differentiation: most teams stop at a road mask; you deliver topology + a resilience simulator = decision-support, not a segmentation demo. Aligns to ISRO NNRMS.

**Risk to manage:** Lowest conceptual ambiguity of the three; just execute the 4 phases and nail the dashboard.

### [2nd] SILVER PS12 — Satellite Frame Interpolation (Optical Flow)

Data: fully open | Best wow-animation demo | Pretrained models exist

**Idea in one line:** Turn INSAT 30-min frames into smooth 15-/7.5-min "video" by AI-interpolating intermediate frames — sharper near-real-time cyclone/storm/fire tracking with a side-by-side time-lapse dashboard.

- Why it wins: before/after time-lapse is an instant visual sell; clean objective metrics (SSIM/PSNR/MSE/FSIM).
- Why it fits you: fine-tune pretrained RIFE/Super-SloMo (fast) + build the web visualization — light on data wrangling, heavy on product polish.
- Data (highest): GOES-19 ABI open on AWS, Himawari open, INSAT-3DS/3DR TIR via MOSDAC. Train on GOES, apply to INSAT.

- Differentiation: satellite-specific optical flow tuned to cloud motion vs generic video models that blur fast cloud dynamics.

**Risk to manage:** Lowest data risk; main effort is a clean .nc pipeline + a crisp metrics story. The safe-but-impressive pick.

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### [3rd] BRONZE PS5 — AI-Powered Digital Twin of India Climate

Data: open, some parsing | Biggest vision/narrative | Atmanirbhar / national impact

**Idea in one line:** A living virtual replica of India climate that ingests IMD + INSAT data, predicts rainfall/temperature for a pilot region, and runs what-if scenarios on an interactive map dashboard.

- Why it wins: the "Digital Twin + Atmanirbhar Bharat" story is the most quotable, mission-aligned narrative in the set.
- Why it fits you: map dashboard + what-if simulator + ML PoC = product-heavy, your strength; pilot-region scope keeps it feasible.
- Data (good): IMD gridded rainfall/temp open (binary .grd — needs a parser), INSAT via MOSDAC (free account). Budget a day for IMD format.
- Differentiation: frame it as a true twin (live state + scenario engine), not just a forecast model.

**Risk to manage:** "Digital twin" is broad — judges punish vagueness. Win by scoping hard to one basin + two variables + a concrete what-if.

My recommendation: PS4 for the highest ceiling (novel, demo-rich, data-ready, on your skills); PS12 for the lowest-risk dazzling demo; PS5 for the biggest national-mission story. Tell me which and I build the full 10-slide deck + architecture + dashboard mock.

## 6 · The plan — 6 days now, ~16 days if selected

### Phase A · Idea sprint (25 Jun -> 1 Jul)

|                |                                                                                                                       |
|----------------|-----------------------------------------------------------------------------------------------------------------------|
| Day 1 (today)  | Lock the PS (PS4/PS12/PS5). I draft the full deck + architecture + flow diagram.                                      |
| Day 2–3        | Sharpen slide 3 (USP). Build dashboard wireframe (slide 6) + architecture (slide 7). Verify every dataset link opens. |
| Day 4          | Tech stack + optional cost slide; tighten copy; add the 2 visuals judges love.                                        |
| Day 5 (30 Jun) | Internal review, proofread, export to official template, SUBMIT a day early.                                          |
| 1 Jul          | Buffer / final submit. Deadline 11:59 PM IST.                                                                         |

Phase B · The wait (1–20 Jul): skim explainer recordings, set up data accounts (Bhoonidhi, MOSDAC, NOAA AWS), stub the repo.

### Phase C · Build runway IF selected (21 Jul -> 6 Aug ~ 16 days)

Your real build window. Week 1: data pipeline + baseline model (e.g., PS4: U-Net/transformer road segmentation on SpaceNet/DeepGlobe, OSM ground-truth automation, graph-healing). Week 2: dashboard skeleton + integration + a rehearsed end-to-end demo. Pre-build the heavy parts so the 30 finale hours go to wow-factor, not setup.

Phase D · Grand Finale (6–7 Aug, 30 hrs): drop in finale-only data (Cartosat-3 for PS4), finalize the model, polish the interactive demo, nail the presentation.

## 7 · Full problem-statement compendium (all 15)

Faithful summary of each official problem statement. Official images are in the companion HTML.

### PS1 Urban Heat Mitigation & Cooling via AI/ML

**Data readiness:** High (open)

Physics-informed geospatial AI to find urban heat hotspots, quantify drivers (land use, morphology, vegetation, atmosphere), model LST, and simulate cooling interventions (green/cool roofs, albedo, trees, water) with estimated °C reduction per option.

**Datasets / eval:** Landsat-8 & ECOSTRESS LST, Sentinel-2 LULC, ERA5 + CPCB, OSM/GHSL. Tools: SOLWEIG, InVEST.

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### PS2 Generative-AI Cloud Removal & Reconstruction (LISS-IV)

**Data readiness:** Medium (LISS-IV pairs tricky)

GenAI framework to remove clouds and reconstruct surface in LISS-IV using spatial+spectral+temporal cues; explore diffusion/GAN/transformer/multimodal fusion; output cloud-free analysis-ready products.

**Datasets / eval:** LISS-IV cloudy/clear (Bhoonidhi); optional Sentinel-1 SAR, Sentinel-2, DEM.

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### PS3 Surface AQI & HCHO Hotspots over India

**Data readiness:** High (open via GEE)

Obj-1: satellite-based surface AQI over India (CNN/LSTM fusing satellite + CPCB + met). Obj-2: HCHO hotspots in biomass-burning seasons, source regions (Indo-Gangetic Plain), fire-to-HCHO transport.

**Datasets / eval:** INSAT-3D AOD (MOSDAC), Sentinel-5P TROPOMI, CPCB, MODIS/VIIRS fire (FIRMS), ERA5/MERRA-2. Eval: RMSE/R/MAE + hotspot clarity.

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### PS4 Route Resilience: Road Extraction + Graph Criticality [TOP PICK]

**Data readiness:** High (open + auto OSM GT)

Transformer model infers road continuity under occlusion; heal masks into a routable graph (MST + Disjoint-Set); find Gatekeeper Nodes (betweenness); node-ablation stress test -> Resilience Index; interactive dashboard.

**Datasets / eval:** Sentinel-2, Resourcesat LISS-IV, SpaceNet/DeepGlobe/OpenSatMap, OSM; Cartosat-3 at finale. Eval: IoU/Dice, occlusion-recall, connectivity ratio, topology vs OSM.

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### PS5 AI Digital Twin of India Climate [SHORTLIST]

**Data readiness:** Good (IMD binary parse)

High-fidelity virtual replica fusing satellite + IMD + reanalysis; near-real-time states + short-term rainfall/temp prediction for a pilot region; interactive map dashboard + what-if scenario module.

**Datasets / eval:** IMD gridded rainfall/temp, INSAT LST/SST/rainfall (MOSDAC), Bhuvan/NICES. Strong Atmanirbhar narrative.

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### PS6 Crop Type + Moisture Stress + Irrigation Advisory

**Data readiness:** Medium (ground-truth labels)

Automated crop-type mapping (multi-temporal), phenology-aware moisture-stress detection, and 8-day crop-water-deficit -> irrigation advisory for canal command areas. Supports PMKSY/PMFBY.

**Datasets / eval:** LISS-III/AWiFS/Sentinel-2/Landsat/MODIS optical, EOS-04/Sentinel-1 SAR, rainfall/ET, crop ground truth. Eval: >85% acc + Kappa.

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### PS7 Exoplanet Detection from Noisy Light Curves

**Data readiness:** High (TESS open + curated)

Pipeline to find periodic dips in noisy crowded-field light curves and classify transit vs eclipsing-binary vs blend; report SNR; fit transit depth/period/duration. <=3-page report.

**Datasets / eval:** TESS raw light curves (MAST) + provided curated training set. Note: NASA data, less ISRO-specific.

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## PS8 Lunar Subsurface Ice (Chandrayaan-2) [DATA AT FINALE]

**Data readiness:** Low (data only at finale)

Map subsurface-ice doubly-shadowed craters via radar polarimetry ( $CPR > 1$ ,  $DOP < 0.13$ ); propose a safe landing site; design rover traverse; estimate ice volume top ~5 m. Idea phase = methodology.

**Datasets / eval:** Chandrayaan-2 DFSAR + OHRC, supplied during the 30-hr finale. Highest prestige, biggest data risk.

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## PS9 Wavefront Reconstruction (SH-WFS)

**Data readiness:** Provided (niche)

From Shack-Hartmann spot fields: centroid spots, reconstruct wavefront phase maps, derive Fried parameter  $r_0$  & coherence time  $\tau_0$ , compute deformable-mirror actuator maps (Fried geometry, coupling), fast for ~10 ms turbulence.

**Datasets / eval:** SH-WFS .bmp frames + MLA/DM parameters (provided). C-level real-time DSP; far from product strengths.

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## PS10 Infrared Image Colorization & Enhancement

**Data readiness:** High (Landsat open)

End-to-end pipeline that super-resolves/sharpens IR satellite imagery and colorizes IR->RGB preserving semantics (water->blue), boosting downstream detection.

**Datasets / eval:** Landsat-8/9 thermal+IR & RGB (USGS). CycleGAN/Pix2Pix. Eval: PSNR/SSIM/FID, inference time, no hallucinations.

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## PS11 Cross-Modal Satellite Image Retrieval

**Data readiness:** Medium (aligning modalities)

Shared embedding space for same-modal and cross-modal retrieval (optical<->SAR<->MS); ranked top-5/top-10, low per-query time. Pretrained/foundation models allowed.

**Datasets / eval:** Multi-sensor RS (optical/MS/SAR) + optional labels. Eval: F1@5/@10 same & cross modal, retrieval time.

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## PS12 Satellite Frame Interpolation (Optical Flow) [SHORTLIST]

**Data readiness:** Highest (GOES open on AWS)

Optical-flow + DL frame interpolation to lift INSAT temporal resolution 30->15->7.5 min for sharper cyclone/storm/fire tracking; dashboard with original vs interpolated time-lapse; .nc I/O.

**Datasets / eval:** GOES-19 ABI (NOAA AWS), INSAT-3DS/3DR TIR (MOSDAC), Himawari-8. Eval: SSIM/PSNR/MSE/FSIM + web GUI.

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## PS13 Air-Gapped Predictive MPLS Copilot

**Data readiness:** Self-generated (controllable)

Offline air-gapped AI NOC copilot: predict faults before impact (time-series + graph anomaly) and explain via a self-hosted quantized LLM (Mistral/LLaMA-3/Phi-3) with RAG over runbooks; zero outbound.

**Datasets / eval:** Simulated SD-WAN/MPLS telemetry (EVE-NG/GNS3/Containerlab). Eval (weighted): Tech 35% / Copilot 35% / Air-gap 20% / Docs 10%. Heaviest infra.

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## PS14 Forecast GEO Particle Radiation

**Data readiness:** Provided (11 yrs)

Forecast geostationary electron flux 30-45 min ahead (+6h/12h) from solar-wind drivers; read/process/visualize CDF; build/tune/validate an AI/ML time-series model.

**Datasets / eval:** GOES >2 MeV flux (11 yr), Wind solar-wind (11 yr), ISRO GRASP/GSAT (1-2 yr) in CDF, provided.

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## PS15 Solar Flare Nowcast/Forecast (Aditya-L1)

**Data readiness:** Medium (ISSDC-PRADAN)

Combine soft+hard X-ray light curves to nowcast (detect/classify) flares and forecast from precursors with lead time; alert-visualization interface.

**Datasets / eval:** Aditya-L1 SoLEXS (L1) + HEL1OS (L1) via ISRO ISSDC PRADAN. Eval: low/high-class detection, high TPR/low FAR, lead time.

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Sources: [hack2skill.com/event/bah2026](https://hack2skill.com/event/bah2026) (official site & team dashboard), [isro.gov.in](https://isro.gov.in). Prepared for Team Syntax Error, 25 Jun 2026.